

Retail Sales Analysis Using Python

Objective

The objective of this project was to analyse retail sales data to identify sales trends, top-performing categories, and regional performance. The analysis aimed to provide business insights that could support decision making and improve sales performance.

Dataset

The dataset contained retail transaction data including Order Date, Sales, Profit, Category, Segment, and Region.

Tools Used

- Python
- Pandas
- Matplotlib
- VS Code
- CSV Dataset

Data Cleaning

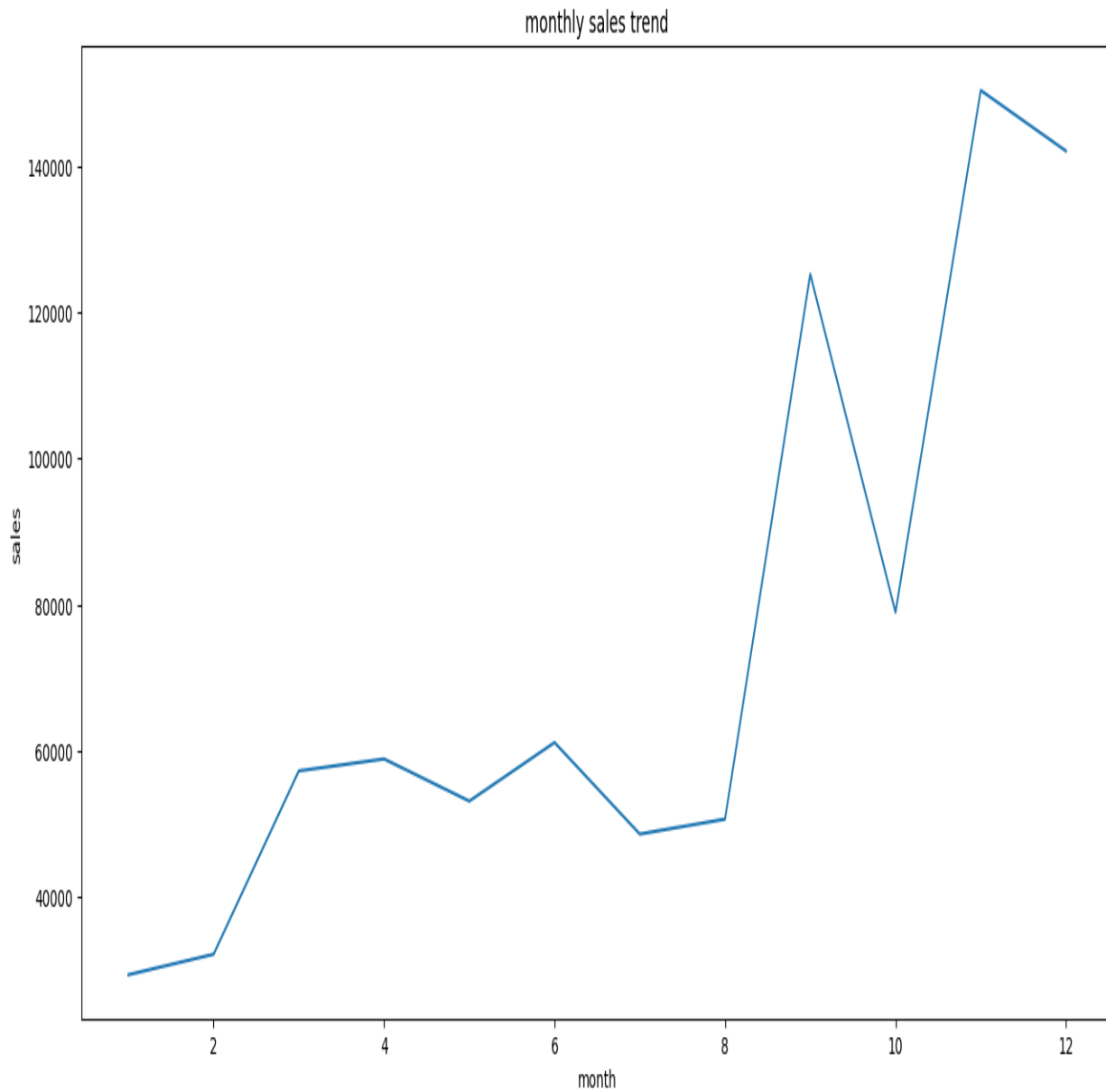
- Imported CSV data using pandas
- Converted date columns to datetime format
- Checked for missing values; no missing values were identified.
- Checked for duplicate records; no duplicate records were identified.
- Inspected column names and data types to verify data was correctly loaded and suitable for analysis.

Analysis Performed

Monthly Sales Trend

- Extracted month from date column and analysed Monthly sales performance

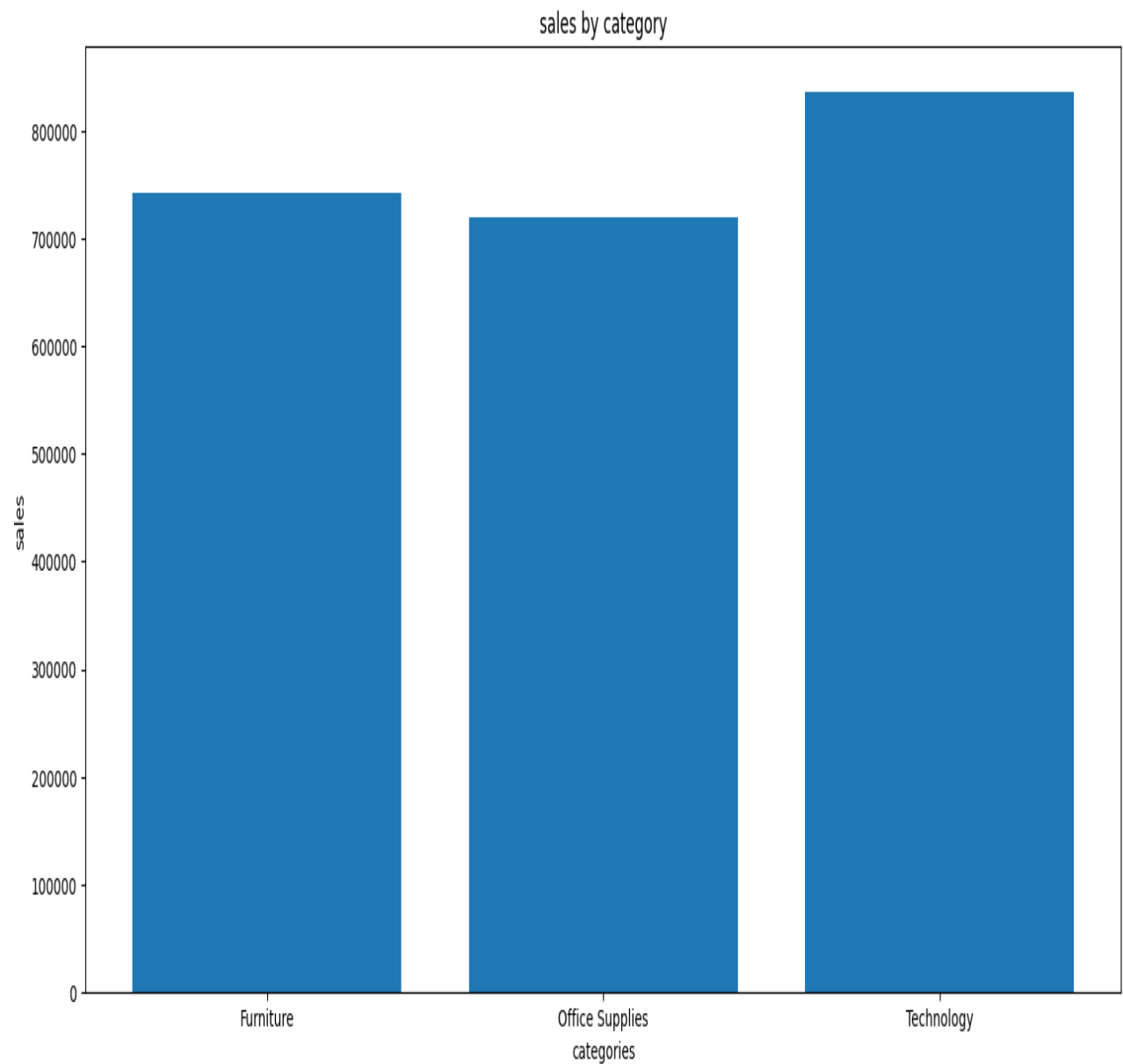
- Figure 1: Monthly sales trend



- **Insight:** Sales showed a fluctuating trend over time indicating, inconsistent monthly performance and opportunity for improved planning.

Category Analysis

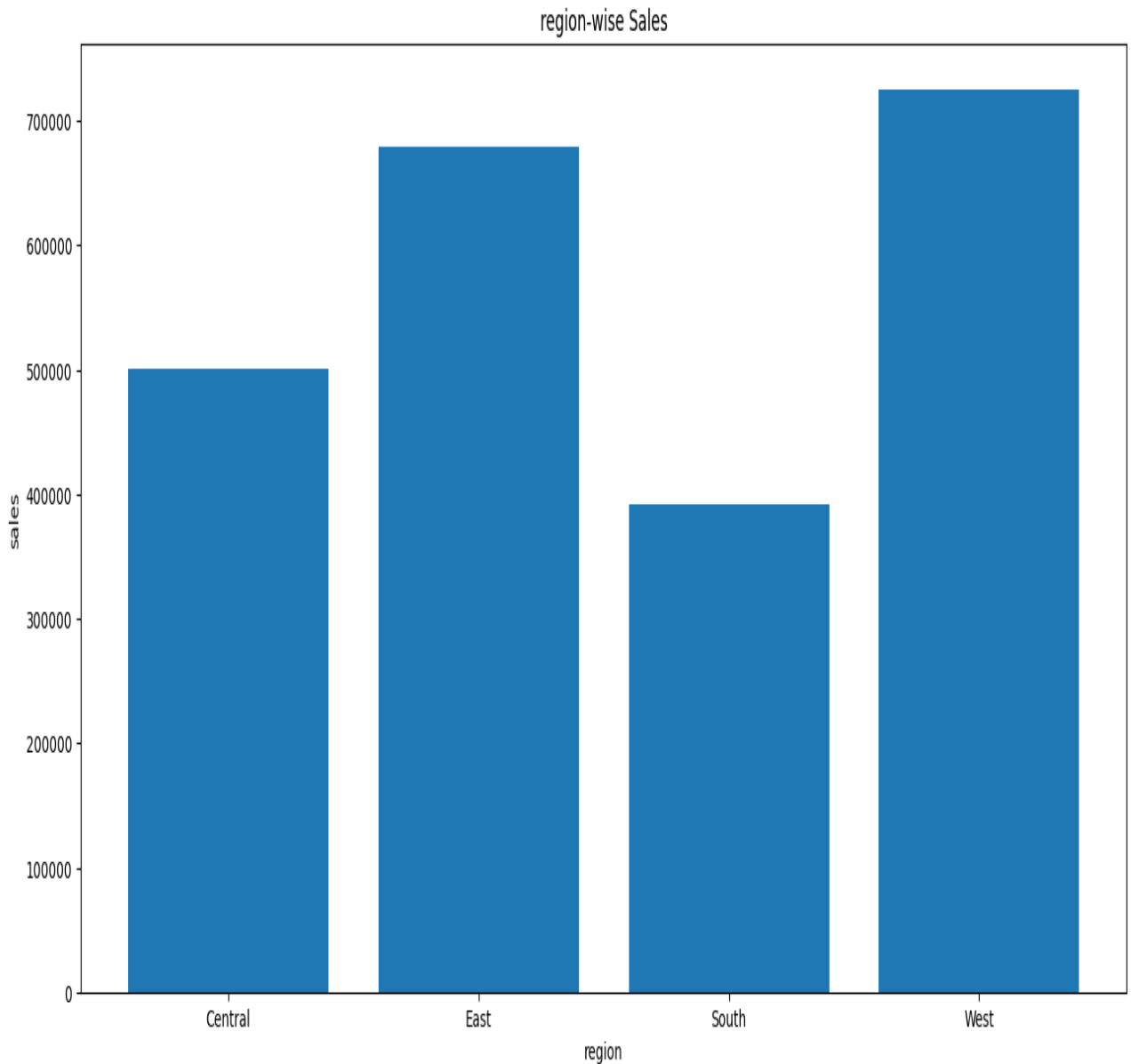
- Analysed sales contribution by category
- Figure 2: Sales by category



- **Insight:** Technology generates the highest share of revenue among all categories, indicating strong customer demand and business performance.

Regional Analysis

- Compared sales performance across regions
- Figure 3: Region-wise sales



- **Insight:** South region recorded lowest sales contribution, indicating low performance and highlighting an opportunity for growth.

High Sales Orders

- Filtered high-value transactions
- Exported filtered results

Key Insights

- Sales showed a fluctuating trend over time.
- Technology generated the highest revenue.
- The South region contributed the lowest share of revenue.

Recommendations

- Increase focus on high-performing technology products.
- Improve sales strategy in underperforming regions.
- Monitor sales trends regularly to support future planning and business Decision making.
- Investigate factors contributing to lower sales in south region.

Project Summary

- This project demonstrated the use of python, pandas, and matplotlib to Clean, analyse and visualize retail sales data. Through trend, category, and regional analysis, actionable business insights were identified to support data-driven decision making.